AUBAID AHMED SAIYED

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Summary

Software Engineer with expertise in Java, Python, JavaScript, Node.js, React, and REST API design. Strong grounding in data structures, algorithms, and scalable backend architecture.

Experience

Social Summer of Code'26 May 2026 – June 2026
Contributor *Remote*

- Eliminated file-not-found failures by **building a centralised ingestion layer** with job-scoped deterministic storage and streaming uploads, ensuring reliable file access across all pipeline stages.
- Connected an existing 6-stage processing pipeline (OCR -> Vision -> Source Lookup -> Tagging -> Storage -> Indexing) through a background job worker, **enabling fully asynchronous execution of incoming uploads**.
- **Improved pipeline fault tolerance and observability**, by adding per-stage error isolation, lifecycle tracking, and detailed stage-level timing logs with REST-based job status APIs for debugging and monitoring.

Projects

Vector AI | Tech Stack: MongoDB - Node.js - Express.js - React.js - Python | Github | Link

- Built a **retail demand intelligence platform** using React, Express.js, MongoDB Atlas, FastAPI, Redis, and XGBoost for demand forecasting, inventory analytics, and EOQ-based inventory optimization.
- Developed and benchmarked **XGBoost forecasting models**, achieving **17% MAPE** through structured backtesting.
- Reduced forecast API latency by **75%**, from **792ms to 193ms**, using Redis caching, MongoDB indexing, and request-level caching.
- Accelerated store-wide forecasting by **87%**, from **6.85s to 0.9s**, through parallel execution and SKU-level workload partitioning.
- Improved concurrent workload performance by **78%**, reducing average latency from **756ms to 166ms** and p95 latency from 950ms to 209ms, through query optimization and request deduplication.

StackLens | Tech Stack: JavaScript - React.js - Node.js - Express.js - Babel (AST) - Dagre | Github | Link

- Architected a **stateless static analysis engine** using Node.js/React that **parses ASTs across 1,400+ file repositories**, processing code entirely in memory to **eliminate storage overhead while enabling scalable concurrent analysis**.
- Engineered **O(1) adjacency list construction using hash-based indexing**, removing O(N^2) path-resolution patterns and **reducing graph construction overhead to approximately 2ms** in measured runs.
- Implemented **O(V+E) BFS/DFS-based graph algorithms with Set-based operations to compute dependency impact paths, detect cycles**, and identify orphaned modules in large-scale codebases.

Safe Stay | Tech Stack: MongoDB - Node.js - Express.js - React.js | Github | Link

- Developed Safe Stay, a **MERN stack emergency shelter platform** connecting evacuees with temporary, verified housing during crises and disasters.
- Implemented secure **JWT authentication**, RESTful APIs, and backend services using Node.js and Express.js.
- Designed a **scalable backend architecture** with middleware, file uploads, error handling, and deployment-ready environment configuration.
- Built a responsive frontend interface using React, Vite, and Tailwind CSS with role-based dashboards for hosts, evacuees, and admins.

Skills

Languages: Python, Java, JavaScript, HTML, CSS
Frameworks & Technologies: React.js, Node.js, Express.js, Tailwind CSS, RESTful APIs
Coursework: Data Structures & Algorithms, Database Management Systems, Object-Oriented Programming, Computer Networks
Tools/Techniques: MySQL, Git/GitHub, MongoDB, Product Thinking
Soft Skills: Communication, Teamwork, Leadership

Education

Charotar University of Science and Technology, Anand, India 2024 – 2028
B.Tech in Information Technology

Kendriya Vidhyalaya Rajkot, Rajkot, India 2023 – 2024
Class 12, Grade: 93%